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Original Research

Comparison of Surgical Techniques and Joint Injections for Base of Thumb Osteoarthritis: A Systematic Review



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Purpose: To perform an updated systematic review investigating the various surgical and injection interventions for thumb carpometacarpal (CMC) joint arthritis.

Methods: A systematic literature search was conducted in MEDLINE, Embase, Web of Science, and Cochrane databases from inception to April 2024 to identify prospective, randomized studies comparing surgical and/or minimally invasive interventions for thumb CMC joint osteoarthritis with a minimum follow-up of 12 months. Reference lists of previous reviews were screened to identify additional publications. Qualitative analysis was performed for primary and secondary outcomes based on the direction of effect (statistically higher, lower, or no difference) because of marked heterogeneity of data and inadequate statistical power. Cochrane's risk of bias 2 tool assessed the quality of included studies.

Results: Seventeen studies encompassing 1,166 thumbs were included in this review. Fourteen studies compared surgical interventions, whereas three studies compared various intra-articular injections. Five studies had a low risk of bias, eight had a moderate risk of bias, and four had a high risk of bias. Regarding primary outcomes, there was no difference in function between trapeziectomy and arthroplasty, key pinch strength between partial and total trapeziectomy, and pain and function between trapeziectomy and arthrodesis. Arthrodesis was associated with a higher complication rate compared with trapeziectomy. Platelet-rich plasma injections had improved pain, function, and patient satisfaction outcomes compared with corticosteroid injections.

Conclusions: No surgical intervention was superior to the other for treating the first CMC joint osteoarthritis. Trapeziectomy remains the simplest and one of the most established surgical modalities with good long-term outcomes and acceptable side effect profiles. Platelet-rich plasma injections may be a useful alternative to corticosteroids in first CMC osteoarthritis.

Type of study/level of evidence: Therapeutic IB.

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Thumb carpometacarpal (CMC) joint osteoarthritis, also known as base of thumb arthritis or trapeziometacarpal arthritis, is a prevalent condition that significantly impairs hand function and quality of life, particularly among the elderly and postmenopausal women.¹ The CMC joint is crucial for thumb opposition and pinching, and its degeneration can lead to severe pain, functional

limitations, and disability.² As the population ages, the incidence of this debilitating condition is expected to rise, highlighting the need for effective management strategies.¹

Conservative management of CMC joint osteoarthritis typically includes nonsteroidal anti-inflammatory drugs, corticosteroid injections, and splinting, which aim to alleviate pain and inflammation but often provide only temporary relief.^{3,4} When conservative treatments fail, surgical interventions may become necessary. The traditional surgical approach has been trapeziectomy, with or without ligament reconstruction and tendon interposition (LRTI), which has long been considered the standard of care.^{5,6} This

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procedure involves removing the trapezium and stabilizing the thumb metacarpal using harvested tendons, which purportedly helps reduce pain and improves functional efficiency.⁵ However, newer surgical techniques, such as arthroscopy, osteotomy, and various forms of arthroplasty, have emerged. These techniques aim to provide similar or better outcomes with potentially fewer complications and faster recovery.⁷

Joint injections are alternatives to traditional surgery.⁸ These include intra-articular injections of substances like hyaluronic acid, platelet-rich plasma (PRP), or corticosteroids. Platelet-rich plasma and fat grafting have shown promise in early studies, demonstrating potential benefits in pain reduction and functional improvement because of their regenerative properties.⁹ Despite the growing interest, the effectiveness and long-term outcomes of these minimally invasive treatments compared with traditional surgical options still need to be determined.

The existing literature presents a mixed picture regarding the comparative effectiveness of different surgical and minimally invasive interventions for CMC joint osteoarthritis. Studies have shown variable outcomes about pain relief, functional improvement, and complication rates, making it difficult for clinicians to determine the most appropriate treatment for their patients.¹⁰ Moreover, the lack of standardized reporting and variability in study designs further complicates the evidence synthesis. Since the previous systematic review on the topic more than a decade ago, there have been a plethora of studies comparing various interventions for thumb CMC joint osteoarthritis.¹¹ Therefore, the present study aimed to perform an updated systematic review comparing outcomes of various surgical techniques and minimally invasive interventions in patients with thumb CMC joint arthritis with a follow-up of at least 12 months.

Methods

This review followed the preferred reporting items for systematic reviews and meta-analyses guidelines and was registered on the International Prospective Register of Systematic Reviews. Two reviewers (S.B. and L.W.) independently performed the title and abstract screen, full-text review of eligible studies, data extraction, and risk of bias assessment. Discrepancies were resolved by consensus or with a third reviewer's involvement if required.

Search strategy

A systematic literature search was conducted across MEDLINE, Embase, Web of Science, and Cochrane databases from inception to 26 April 2024 with no language restrictions using the following search terms in the Title and Abstract fields: ("thumb carpometacarpal joint" or "first carpometacarpal joint" or "trapeziometacarpal joint") and ("osteoarthritis" or "arthritis" or "rhizarthrosis") and ("surgery" or "surgical intervention*" or "minimally invasive intervention*" or "treatment" or "therap*" or "arthrodesis" or "arthroplasty" or "trapeziectomy" or "ligament reconstruction" or "LRTI" or "steroid injection" or "CSI" or "corticosteroid injection" or "PRP" or "platelet-rich plasma" or "hyaluronic acid" or "denervation" or "osteotomy" or "prolotherapy"). Reference lists of previous reviews were screened to identify additional relevant publications.

Eligibility criteria

Prospective, parallel-arm, randomized studies (blinded and unblinded) comparing different surgical and/or minimally invasive interventions for thumb CMC joint osteoarthritis with a minimum follow-up of 12 months were eligible for inclusion. Interventions of

interest included arthrodesis, arthroplasty, trapeziectomy with or without soft tissue reconstruction, denervation, osteotomy, and steroid, hyaluronic acid, or PRP injections. In this study, arthroplasty refers to joint replacement with prosthetic implants. Retrospective studies, nonrandomized prospective studies, narrative reviews, case reports, case series, studies with no comparator groups, studies investigating only noninvasive interventions such as hand or physical therapy, studies with noninvasive interventions as control groups, studies comparing different implants or techniques (eg, cemented versus uncemented arthroplasty, different spacer devices, and various tendon interposition techniques) instead of broad surgical categories (eg, trapeziectomy, arthroplasty, denervation, etc) were excluded.

Outcomes

The following outcomes were investigated as reported in individual studies: pain, functional outcomes, range of motion, strength, and complications. Cochrane's risk of bias 2 tool assessed the quality of included studies.

Data analysis

Because of limited data available for some secondary outcomes across studies, the decision was not to pool results in a meta-analysis as any such analysis would be fundamentally underpowered. Qualitative analysis was performed for primary and secondary outcomes based on the direction of effect (statistically higher, lower, or no difference). Clinically significant differences in primary outcomes were considered clinically relevant if there was a difference of at least 15 points in the visual analog scale for pain, 15 points in disabilities of the arm, shoulder, and hand score, and 0.5 kg for key pinch strength, as used previously.¹²

Results

Of 1,321 unique hits identified by our search, 17 studies encompassing 1,166 thumbs were included in this review (Fig. 1; Table 1).^{6,9,13–26} Fourteen studies compared surgical interventions, whereas three studies compared minimally invasive techniques.^{6,9,13–27} Of the surgical interventions, six compared trapeziectomy to trapeziectomy with LRTI; three studies compared trapeziectomy with arthroplasty; two compared arthrodesis and trapeziectomy; one compared partial trapeziectomy to total trapeziectomy; one compared trapeziectomy and LR with trapeziectomy with LRTI; and one compared trapeziectomy to denervation.^{6,14–16,18–27} Regarding minimally invasive interventions, one study compared autologous fat with PRP injections; one corticosteroid with hyaluronic acid injections; and one PRP with corticosteroid injections.^{9,13,17}

Risk of bias

Table 2 summarizes the risk of bias assessment for included studies. Overall, five studies were deemed to have a low risk of bias.^{9,14,17,20,21} There were some methodological concerns in eight studies, stemming mainly from issues in the "intervention deviation" domain. The remaining four studies had a high risk of bias, primarily because of bias in the "result selection" and "intervention deviation" domains.^{6,13,15,16,18,19,22–27}

Outcomes

Only five studies reported primary outcomes: pain, function, and key pinch strength.^{9,14,20–22} Of these, the study by Vermeulen

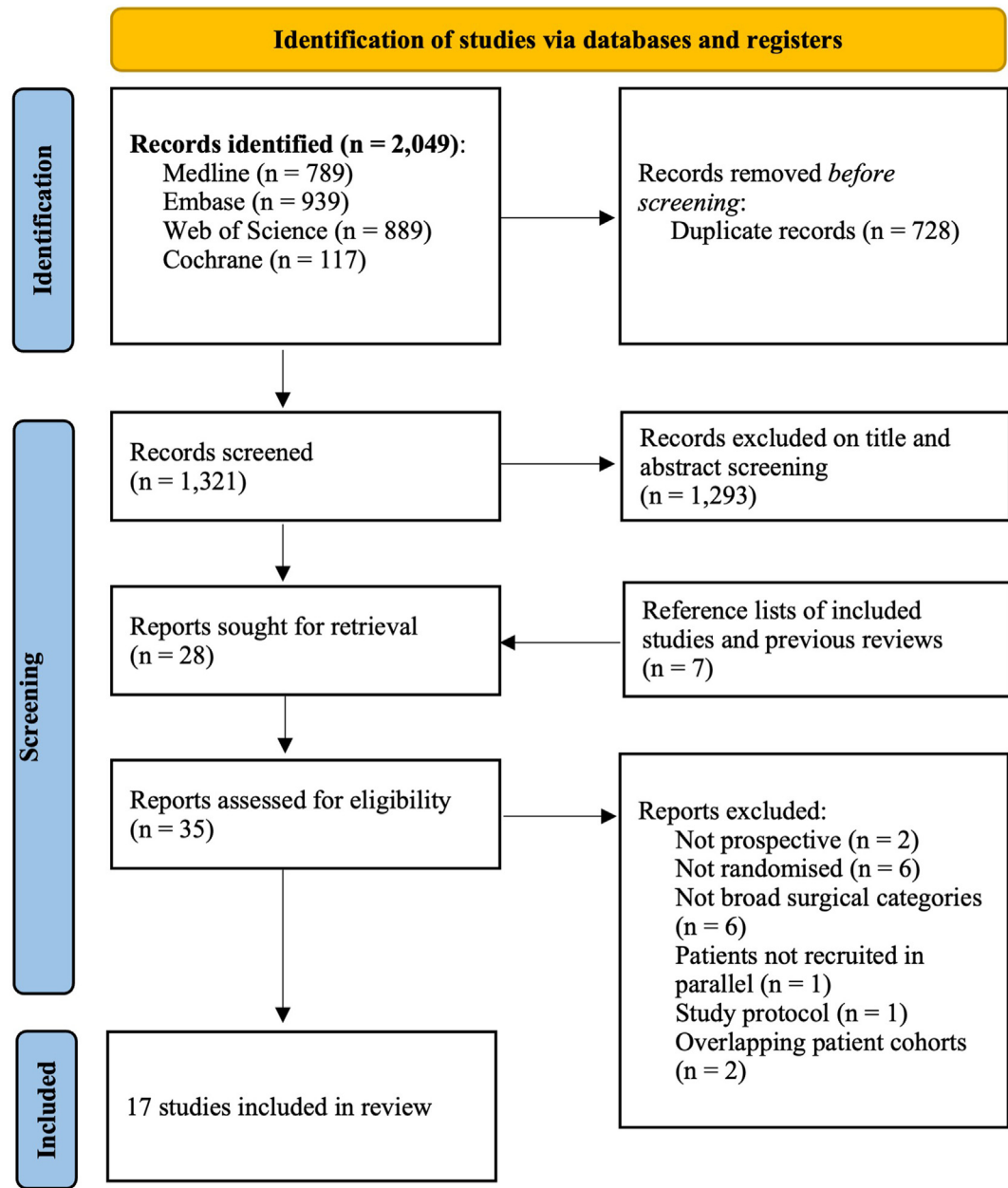


Figure 1. Flow diagram outlining the screening and exclusion process.

et al,²² was terminated prematurely owing to higher complications in the arthrodesis group. These include, among others, nonunion requiring additional surgery (n = 2) and complex regional pain syndrome (n = 1). Therefore, results from only four studies, which also had the lowest risk of bias, were adequately powered to compare interventions. However, the data needed to be more amenable to pooling in a meta-analysis due to disparate interventions studied and measured outcomes, leading to a qualitative synthesis (Table 3).^{6,9,13–27} There was no difference in long-term function between trapeziectomy and arthroplasty, key pinch strength between partial and total trapeziectomy, and pain and function between trapeziectomy and arthrodesis.^{14,20–22} Patients undergoing arthrodesis experienced more complications such as delayed union and nonunion compared with the trapeziectomy group.²² No difference in pain or function was observed between trapeziectomy and denervation.¹⁹ Changes in secondary outcomes are illustrated in Table 3.^{6,9,13–27}

Discussion

This systematic review found no surgical intervention superior to another in managing thumb CMC joint osteoarthritis. The study's strengths include rigorous inclusion criteria that selected randomized studies with relatively long follow-up periods, thereby allowing better determination of an intervention's longevity. We found no difference in outcomes between trapeziectomy and arthroplasty. Although this is consistent with most existing literature, some suggest superior key pinch strength and thumb opposition with arthroplasty.^{4,11,12,28–32} However, these latter reviews included retrospective and nonrandomized prospective studies with significant heterogeneity in their pooled analyses, necessitating caution in their interpretation. Arthroplasty may be associated with better grip strength and thumb movement; however, any such benefits must be weighed against higher complication and revision surgery rate compared with trapeziectomy.^{32,33} A

Table 1
Characteristics of Included Studies

Study	Study Period	n Thumbs	Loss to Follow-Up (%)	Mean Age (y)	% F	Follow-Up (mo)	Blinding	Eaton-Littler Stage	Grade I (%)	Grade II (%)	Grade III (%)	Grade IV (%)	Exclusion Criteria	Interventions Compared
de Jong et al ¹⁴	2014–2018	62	3	60	100	12	Double	II–III	0	25	37	0	Previous operations for CMC joint OA, posttraumatic OA, RA, SLE, gout, STT arthrosis, CTS, de Quervain tenosynovitis, neurological disorders affecting the hand	1. Trapeziectomy 2. Arthroplasty
Guzzini et al ¹⁵	2017–2020	150	2	67	81	24	Single	II–III	-	-	-	-	STT arthrosis, poor bone quality radiologically, RA	1. Trapeziectomy + TI 2. Arthroplasty
Thorkildsen et al ²¹	2008–2016	40	10	62.5	70	24	Single	II–IV	0	2.5	95	2.5	STT arthrosis, large trapezial cysts, injuries of the thumb or carpus	1. Trapeziectomy + LRTI 2. Arthroplasty
Belcher and Nicholl ²⁵	1996–1998	43	3	60	88	14	Unblinded	-	-	-	-	-	RA, patient undergoing another procedure at the same time	1. Trapeziectomy 2. Trapeziectomy + LRTI
Brennan et al ⁶	2001–2003	65	43	76	79	210	-	III–IV	-	-	-	-	-	1. Trapeziectomy 2. Trapeziectomy + LRTI
Field and Buchanan ²⁴	2001–2003	65	0	55	86	12	-	III–IV	0	0	45	55	-	1. Trapeziectomy 2. Trapeziectomy + LRTI
Gangopadhyay et al ²⁶	1992–2001	153	5.6	57	100	Minimum 72	Unblinded	II–IV	-	-	-	-	-	1. Trapeziectomy 2. Trapeziectomy + TI 3. Trapeziectomy + LRTI
Hart et al ²⁷	1995–2002	40	0	59	65	81.6	Single	IV	0	0	0	100	RA, posttraumatic arthritis, previous thumb operation, coexistent hand pathology	1. Trapeziectomy + LRTI 2. Arthrodesis
Kriegs-Au et al ¹⁶	1995–1998	52	28	59	81	48.2	Single	II–IV	-	16	71	13	RA, posttraumatic arthritis, previous thumb operation, coexistent hand pathology	1. Trapeziectomy + LR 2. Trapeziectomy + LRTI
Salem and Davis ¹⁸	2002–2005	131	11	-	85	72	Unblinded	-	-	-	-	-	-	1. Trapeziectomy 2. Trapeziectomy + LRTI
Zarezadeh et al ²³	2019–2021	70	20	55	68	12	Single	-	-	-	-	-	RA, previous joint surgery	1. Trapeziectomy 2. Trapeziectomy + LRTI
Vermeulen et al ²²	2008–2011	38	0	59	100	12	Single	II–III	0	34	66	0	RA, posttraumatic arthritis, previous thumb surgery	1. Trapeziectomy + LRTI 2. Arthrodesis
Salibi et al ¹⁹	2005–2013	55	18	59	76	60	Unblinded	-	-	-	-	-	Previous base of thumb surgery	1. Trapeziectomy 2. Trapezial denervation
Sánchez-Flò et al ²⁰	2014–2015	34	6	60.5	82	12	Single	II–III	0	27	73	0	Cognitive impairment, posttraumatic OA, MCPJ hyperextension	1. Partial trapeziectomy 2. Total trapeziectomy
Bahadir et al ¹³	-	40	-	62	100	12	Single	II–III	-	-	-	-	Previous fracture, CTS, previous surgery or major trauma to hand, inflammatory systemic diseases, de Quervain tenosynovitis, any previous injection	1. Steroid injection 2. Hyaluronic acid injection
Malahias et al ¹⁷	2012 – 2014	33	3	63	79	12	Single	I-III	-	-	-	-	Systemic rheumatic disease, CTS, de Quervain tenosynovitis, history of gout or pseudogout, bleeding predisposition, previous surgery to the affected thumb, previous injection to the thumb base within past 12 months, no evidence of CMC joint space narrowing on plain radiographs	1. Steroid injection 2. PRP injection
Winter et al ⁹	-	95	6	64	84	24	Single	I-IV	1.1	21	60	18	Pregnancy or lactation, CTS, de Quervain tenosynovitis, STT arthrosis, rheumatoid arthritis, multiple joint OA of the hand not involving the first CMC joint	1. Autologous fat injection 2. PRP injection 3. Autologous fat + PRP injection 4. Saline injection

CMC, carpometacarpal; CTS, carpal tunnel syndrome; OA, osteoarthritis; RA, rheumatoid arthritis; SLE, systemic lupus erythematosus; STT, scapho-trapezium-trapezoid.

Table 2
Risk of Bias Assessment for Included Studies According to Cochrane's Risk of Bias 2 Tool for Randomized Trials.

Study	Domain					Overall Result
	Randomization	Intervention Deviations	Missing Data	Outcome Measurement	Result Selection	
Bahadir et al ¹³	Some concerns	Low	Low	Low	Some concerns	Some concerns
Belcher and Nicholl ²⁵	Low	Some concerns	Low	Low	Some concerns	Some concerns
Brennan et al ⁶	Low	Some concerns	Some concerns	Low	Low	Some concerns
de Jong et al ¹⁴	Low	Low	Low	Low	Low	Low
Field and Buchanan ²⁴	Low	Some concerns	Low	Low	Low	Some concerns
Gangopadhyay et al ²⁶	Low	Some concerns	Some concerns	High	High	High
Guzzini et al ¹⁵	Low	Some concerns	Low	Some concerns	Some concerns	Some concerns
Hart et al ²⁷	Some concerns	Some concerns	Low	Some concerns	High	High
Kriegs-Au et al ¹⁶	Low	Some concerns	Some concerns	Low	Low	Some concerns
Malahias et al ¹⁷	Low	Low	Low	Low	Low	Low
Salem and Davis ¹⁸	Low	Some concerns	Low	Low	Low	Some concerns
Salibi et al ¹⁹	High	High	High	Low	Low	High
Sánchez-Flò et al ²⁰	Low	Low	Low	Low	Low	Low
Thorkildsen and Røkkum ²¹	Low	Low	Low	Low	Low	Low
Vermeulen et al ²²	Low	High	Low	Low	Low	High
Winter et al ⁹	Low	Low	Low	Low	Low	Low
Zareza deh et al ²³	Low	Some concerns	Some concerns	Low	Low	Some concerns

comparative review of complications of trapeziectomy and arthroplasty by Latelise and colleagues, who included all studies reporting a complication following either procedure, irrespective of study design or follow-up, found that 6% of patients undergoing trapeziectomy experienced severe complications such as thumb collapse and metacarpophalangeal hyperextension. By contrast, approximately one in every four patients who underwent arthroplasty experienced a severe complication such as dislocation, loosening, or wear.³³ The associated revision rates were 2% for the trapeziectomy group and 13% for the arthroplasty group.³³ In our study, no difference in complication rate between the two groups was observed. Partial trapeziectomy was introduced to preserve healthy joints and prevent first-ray collapse, maximizing thumb length. However, this joint-sparing technique did not provide any advantage over trapeziectomy 1 year after surgery. Overall, the results from this study and previous reviews suggest comparable clinical outcomes between trapeziectomy and arthroplasty, with a possibly higher complication profile associated with arthroplasty.

The addition of soft tissue reconstruction (TI, LR, or LRTI) to augment trapeziectomy is intended to prevent loss of postoperative strength, thumb shortening, and scaphoid impingement.⁵ Similar to results reported previously, we found no benefit in soft tissue reconstruction besides trapeziectomy for the base of thumb osteoarthritis (OA).^{4,5,11,12,28,30,34} Our analysis of secondary outcomes suggests reduced thumb movement, pinch strength, and functional outcomes and greater complications such as tendon rupture or tethering, sensory loss, instability, and pain with soft tissue augmentation.^{6,16,18,23–25} However, the number of complications in each study needed to be bigger to provide any measure of significance. Taken together, results from this study and previous reviews suggest that soft tissue augmentation to trapeziectomy may not offer additional benefits as initially conceived.

Trapeziometacarpal arthrodesis is thought to provide a relatively stable thumb at the cost of manual dexterity.¹⁰ We found arthrodesis inferior to trapeziectomy in terms of function and range of motion. Consistent with previous studies, arthrodesis was associated with more complications than other surgical modalities. Complications specific to arthrodesis included delayed union or nonunion requiring additional surgery, with nonunion rates of 8% to 21% reported in the literature.¹¹ Although this renders arthrodesis a less useful surgical modality for treating first CMC joint OA in most patients, it may still be helpful in specific patient cohorts, such

as those with hypermobility or Ehlers-Danlos syndrome. Comparative evidence in these groups of patients still needs to be provided.

Injections for first CMC joint osteoarthritis include a range of bioactive substances, such as corticosteroids, hyaluronic acid, and PRP, that can be used for intra-articular injections. No clear algorithm exists for the choice of intra-articular injection type in patients with early to moderate-stage osteoarthritis. We found no difference in outcomes between corticosteroid or hyaluronic acid injections regarding pain, function, or grip strength at 1 year.¹³ On the other hand, patients injected with PRP reported improved function and pain compared with corticosteroid injection, with higher overall satisfaction over a similar time frame.¹⁷ The latter study included an unknown proportion of patients with early-stage arthritis (Eaton-Littler grade I), which may have influenced the differential treatment response because of the high concentration of growth factors and other mediators in PRP that augment the natural inflammatory response, especially during early stages of the disease. Although the combination of PRP and autologous fat resulted in the most significant reduction in pain compared with saline, no direct comparison with other injection modalities besides saline was made.⁹ It is also important to acknowledge the presence of studies that demonstrate no discernible differences in functional outcomes and pain scores between corticosteroid injections and saline placebos, suggesting that intra-articular injections, regardless of substance or dose, may amount to a placebo effect.³⁵ Given the absence of findings of clear superiority among injection types, corticosteroids' affordability, safety, and familiarity position them as pragmatic and viable treatment options for the management of base of thumb arthritis. The most optimum dose and type of corticosteroids remains a poorly explored area with limited capacity for pooled analyses, given short study follow-up periods and heterogeneity of treatment.^{4,35} Furthermore, without strong evidence, avoiding injections altogether or using saline or local anesthetic as low-cost, low-risk alternatives for short-term relief may be reasonable considerations. Future studies with larger cohorts, standardized protocols, and more extended follow-up periods are needed to clarify the role of intra-articular injections and establish clear evidence-based recommendations.

Results from this study must be interpreted in the context of some limitations. First are those attributed to the included studies' intrinsic design, that is, single-blind or unblinded design, single center location, and small sample sizes. For example, due to

Table 3
Summary of Outcomes in Included Studies*

Study	Group 1	Group 2	Group 3	Primary Endpoint	Specific Outcomes				Satisfaction
					Function	Strength	Movement	Pain	
de Jong et al ¹⁴	Trapeziectomy	TJA	-	ND	ND	Improved key pinch strength (1.2kg difference)	Improved flexion (6.7° difference) and decreased extension (10° difference)	-	Decreased (63% vs 88%)
Guzzini et al ¹⁵	Trapeziectomy + TI	TJA	-	-	ND	Improved grip strength (25.2kg vs 32kg), pinch strength (2.5kg vs 4.3 kg), and key pinch strength (5.2kg vs 6.6kg)	ND	ND	-
Thorkildsen and Røkkum ²¹	Trapeziectomy + LRTI	TJA	-	ND	ND	ND	Improved extension (42° vs 47°) and abduction (42° vs 49°)	-	Decreased (95% vs 90%)
Belcher and Nicholl ²⁵	Trapeziectomy	Trapeziectomy + LRTI	-	-	-	ND	-	ND	-
Brennan et al ⁶	Trapeziectomy	Trapeziectomy + LRTI	-	-	ND	ND	Decreased extension (79° vs 71°)	-	-
Field and Buchanan ²⁴	Trapeziectomy	Trapeziectomy + LRTI	-	-	-	ND	Decreased abduction (70° vs 62°)	ND	-
Gangopadhyay et al ²⁶	Trapeziectomy	Trapeziectomy + TI	Trapeziectomy + LRTI	-	-	ND	-	-	-
Hart et al ²⁷	Arthrodesis	Trapeziectomy + LRTI	-	-	-	-	-	-	-
Kriegs-Au et al ¹⁶	Trapeziectomy + LR	Trapeziectomy + LRTI	-	-	-	-	Decreased extension (43° vs 39°) and abduction (42° vs 38°)	-	-
Salem and Davis ¹⁸	Trapeziectomy	Trapeziectomy + LRTI	-	-	ND	ND	-	-	-
Zarezaheh et al ²³	Trapeziectomy	Trapeziectomy + LRTI	-	-	ND	Decreased pinch strength (5.4kg vs 4.7kg)	-	ND	-
Vermeulen et al ²²	Trapeziectomy + LRTI	Arthrodesis	-	ND	ND	ND	ND	-	Decreased (86% vs 53%)
Salibi et al ¹⁹	Trapeziectomy	Denervation	-	-	ND	-	-	ND	-
Sánchez-Flò et al ²⁰	Partial trapeziectomy	Total trapeziectomy	-	ND	ND	ND	-	ND	-
Bahadir et al ¹³	Steroid injection	Hyaluronic acid injection	-	-	ND	ND	-	ND	-
Malahias et al ¹⁷	PRP injection	Steroid injection	-	-	Worse (Q-DASH 20.4 vs 43)	-	-	Worse (VAS 2 vs 6.5)	Decreased (69% vs 12.5%)
Winter et al ⁹	Saline injection	PRP	Autologous fat + PRP	Improved	Improved (Q-DASH reduction 15.9 in group 3 compared with group 1)	-	-	Improved in group 3 compared to group 1	-

DASH, Disabilities of the Arm, Shoulder and Hand score; ND, no difference; VAS, Visual Analog Scale.

* Changes in outcome reflect a change in group 2 relative to group 1 unless stated otherwise.

patients' perception of denervation being a less-invasive procedure, recruitment into the randomized study by Salibi et al.⁹¹ was markedly impaired as patients self-selected into their group of choice. Therefore, the results from this study are subject to significant selection bias and should be interpreted cautiously. Second is the marked heterogeneity of reported results (eg, different pain and function scores and inconsistent primary outcomes) and interventions (eg, type, frequency, and dosage of intra-articular injections, and varied implants used for arthroplasty), and lack of adequately powered studies that precluded pooling of results via a meta-analysis. The moderate or high risk of bias in most studies reflects this. Third, it is imperative to recognize that age and sex play pivotal roles in influencing outcomes such as grip and pinch strength.^{36,37} Additionally, anthropometric characteristics have been shown to impact strength.^{36,38} It needs to be clarified whether studies accounted for these potential confounding factors.

In conclusion, our systematic review reinforces the current paradigm that no surgical intervention was superior to the other for treating the first CMC joint OA. Trapeziectomy remains the simplest and one of the most established surgical modalities with good long-term outcomes and acceptable side effect profiles. Corticosteroids remain the most established and readily accessible intra-articular. However, PRP injections are a promising alternative to corticosteroid injections, demonstrating better function and pain outcomes.

Conflicts of Interest

No benefits in any form have been received or will be received related directly to this article.

References

- Teunissen JS, Wouters RM, Bierma-Zeinstra SMA, et al. The prevalence, incidence, and progression of radiographic thumb base osteoarthritis in a population-based cohort: the Rotterdam Study. *Osteoarthritis Cartilage*. 2022;30(4):578–585.
- Zhang Y, Niu J, Kelly-Hayes M, Chaisson CE, Aliabadi P, Felson DT. Prevalence of symptomatic hand osteoarthritis and its impact on functional status among the elderly: the Framingham study. *Am J Epidemiol*. 2002;156(11):1021–1027.
- Spaans AJ, van Minnen LP, Kon M, Schuurman AH, Schreuders AR, Vermeulen GM. Conservative treatment of thumb base osteoarthritis: a systematic review. *J Hand Surg Am*. 2015;40(1):16–21.e1–6.
- Kroon FPB, Carmona L, Schoones JW, Kloppenburg M. Efficacy and safety of non-pharmacological, pharmacological and surgical treatment for hand osteoarthritis: a systematic literature review informing the 2018 update of the EULAR recommendations for the management of hand osteoarthritis. *RMD Open*. 2018;4(2):e000734.
- Li YK, White C, Ignacy TA, Thoma A. Comparison of trapeziectomy and trapeziectomy with ligament reconstruction and tendon interposition: A systematic literature review. *Plast Reconstr Surg*. 2011;128(1):199–207.
- Brennan A, Blackburn J, Thomson J, Field J. Simple trapeziectomy versus trapeziectomy with flexor carpi radialis suspension: a 17-year follow-up of a randomized blind trial. *J Hand Surg Eur Vol*. 2021;46(2):120–124.
- Bellemère P. Pyrocarbon implants for the basal thumb arthritis. *Hand Surg Rehabil*. 2021;40S:S90–S101.
- Kask K, Pivato G, Pozzi A, Susini F, Pegoli L. Minimally invasive procedure for the stabilization of the painful "prearthritic" carpometacarpal joint of the thumb. *Tech Hand Up Extrem Surg*. 2014;18(4):194–198.
- Winter R, Tuca A-C, Justich I, et al. Minimally invasive treatment of trapeziometacarpal osteoarthritis: results of a blinded randomized controlled trial. *Plast Reconstr Surg*. 2023;152(6):1277–1285.
- Seth I, Bulloch G, Seth N, Fogg G, Hunter-Smith DJ, Rozen WM. Efficacy and safety of different trapezium implants for trapeziometacarpal joint osteoarthritis: a systematic review and meta-analysis. *Hand (N Y)*. 2024;19(8):1242–1251.
- Vermeulen GM, Slijper H, Feitz R, Hovius SER, Moojen TM, Selles RW. Surgical management of primary thumb carpometacarpal osteoarthritis: a systematic review. *J Hand Surg Am*. 2011;36(1):157–169.
- Challoumas D, Murray E, Ng N, Putti A, Millar N. A meta-analysis of surgical interventions for base of thumb arthritis. *J Wrist Surg*. 2022;11(6):550–560.
- Bahadir C, Onal B, Dayan VY, Güler N. Comparison of therapeutic effects of sodium hyaluronate and corticosteroid injections on trapeziometacarpal joint osteoarthritis. *Clin Rheumatol*. 2009;28(5):529–533.
- de Jong TR, Bonhof-Jansen EEDJ, Brink SM, de Wildt RP, van Uchelen JH, Werker PMN. Total joint arthroplasty versus trapeziectomy in the treatment of trapeziometacarpal joint arthritis: a randomized controlled trial. *J Hand Surg Eur Vol*. 2023;48(9):884–894.
- Guzzini M, Arioli L, Annibaldi A, Pecchia S, Latini F, Ferretti A. Interposition arthroplasty versus dual cup mobility prosthesis in treatment of trapeziometacarpal joint osteoarthritis: a prospective randomized study. *Hand (N Y)*. 2024;19(8):1260–1268.
- Kriegs-Au G, Petje G, Fojt E, Ganger R, Zachs I. Ligament reconstruction with or without tendon interposition to treat primary thumb carpometacarpal osteoarthritis. A prospective randomized study. *J Bone Joint Surg Am*. 2004;86(2):209–218.
- Malahias MA, Roumeliotis L, Nikolaou VS, Chronopoulos E, Sourlas I, Babis GC. Platelet-rich plasma versus corticosteroid intra-articular injections for the treatment of trapeziometacarpal arthritis: a prospective randomized controlled clinical trial. *Cartilage*. 2021;12(1):51–61.
- Salem H, Davis TRC. Six year outcome excision of the trapezium for trapeziometacarpal joint osteoarthritis: is it improved by ligament reconstruction and temporary Kirschner wire insertion? *J Hand Surg Eur Vol*. 2012;37(3):211–219.
- Salibi A, Hilliam R, Burke FD, Heras-Palou C. Prospective clinical trial comparing trapezial denervation with trapeziectomy for the surgical treatment of arthritis at the base of the thumb. *J Surg Res*. 2019;238:144–151.
- Sánchez-Flò R, Fillat-Gomà F, Marcano-Fernández FA, Berenguer-Sánchez A, Balcells-Nolla P, Torner P. Partial versus total trapeziectomy with interposition arthroplasty for trapeziometacarpal osteoarthritis grade II to III Eaton-Littler: a clinical trial. *J Hand Surg Global Online*. 2020;2(3):133–137.
- Thorkildsen RD, Røkkum M. Trapeziectomy with LRTI or joint replacement for CMC1 arthritis, a randomised controlled trial. *J Plast Surg Hand Surg*. 2019;53(6):361–369.
- Vermeulen GM, Brink SM, Slijper H, et al. Trapeziometacarpal arthrodesis or trapeziectomy with ligament reconstruction in primary trapeziometacarpal osteoarthritis: a randomized controlled trial. *J Bone Jt Surg Am*. 2014;96(9):726–733.
- Zarezaadeh A, Dehghani M, Shahruzian A. Ligament reconstruction and tendon interposition in comparison with hematoma distraction arthroplasty for the treatment of arthritis in the trapeziometacarpal joint. *Int J Burn Trauma*. 2021;11(4):296–303.
- Field J, Buchanan D. To suspend or not to suspend: a randomised single blind trial of simple trapeziectomy versus trapeziectomy and flexor carpi radialis suspension. *J Hand Surg Eur Vol*. 2007;32(4):462–466.
- Belcher HJ, Nicholl JE. A comparison of trapeziectomy with and without ligament reconstruction and tendon interposition. *J Hand Surg Br*. 2000;25(4):350–356.
- Gangopadhyay S, McKenna H, Burke FD, Davis TR. Five- to 18-year follow-up for treatment of trapeziometacarpal osteoarthritis: a prospective comparison of excision, tendon interposition, and ligament reconstruction and tendon interposition. *J Hand Surg Am*. 2012;37(3):411–417.
- Hart R, Janeček M, Šiška V, Kučera B, Štěpčák V. Interposition suspension arthroplasty according to Epping versus arthrodesis for trapeziometacarpal osteoarthritis. *Eur Surg*. 2006;38(6):433–438.
- Saheb RLC, Vaz BAS, Soeira TP, Shimaoka FJ, Herrero CFPDS, Mazzer N. Surgical treatment for rhizarthrosis: a systematic review of the last 10 years. *Acta Ortop Bras*. 2022;30(1):e246704.
- Knightly N, Sullivan P. Surgery for trapeziometacarpal joint osteoarthritis: A systematic review and meta-analysis of recent literature. *J Hand Surg Asian Pac Vol*. 2021;26(2):245–264.
- Martou G, Veltri K, Thoma A. Surgical treatment of osteoarthritis of the carpometacarpal joint of the thumb: a systematic review. *Plast Reconstr Surg*. 2004;114(2):421–432.
- Lee JK, Yoon BH, Lee HI, Kim C, Choi S, Han SH. Prosthetic replacement has a clinical outcome superior to that of trapeziectomy with ligament reconstruction and tendon interposition: a meta-analysis. *Orthopedics*. 2021;44(2):e151–e157.
- Qureshi MK, Halim UA, Khaled AS, Roche SJ, Arshad MS. Trapeziectomy with ligament reconstruction and tendon interposition versus trapeziometacarpal joint replacement for thumb carpometacarpal osteoarthritis: a systematic review and meta-analysis. *J Wrist Surg*. 2022;11(3):272–278.
- Latelise B, Ben Brahim E, Prasil L, Freslon M. Complications of prosthesis versus trapeziectomy in trapeziometacarpal joint arthritis: a systematic review. *Hand Surg Rehabil*. 2024;43(2):101672.
- Rosales RS, Gonzalez-Garcia A, Dorta-Fernandez A, Heras-Palou C. A meta-analysis of the outcomes of ligament reconstruction compared to no reconstruction with or without tendon interposition following trapeziectomy for thumb carpometacarpal joint osteoarthritis. *J Hand Surg Asian Pac Vol*. 2022;27(1):22–31.
- Krez AN, Wu KA, Klifto KM, Pidgeon TS, Klifto CS, Ruch DS. Efficacy of intra-articular corticosteroid injection for nonsurgical management of trapeziometacarpal osteoarthritis: a systematic review and meta-analysis of randomized controlled trials. *J Hand Surg Am Vol*. 2024;49(6):511–525.
- Günther CM, Bürger A, Rickert M, Crispin A, Schulz CU. Grip strength in healthy Caucasian adults: reference values. *J Hand Surg Am*. 2008;33(4):558–565.
- Günther CM, Bürger A, Rickert M, Schulz CU. Key pinch in healthy adults: normative values. *J Hand Surg Eur Vol*. 2008;33(2):144–148.
- Keevil VL, Luben R, Dalzell N, et al. Cross-sectional associations between different measures of obesity and muscle strength in men and women in a British cohort study. *J Nutr Health Aging*. 2015;19(1):3–11.